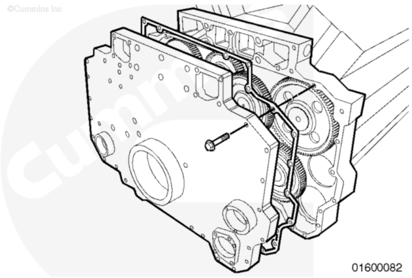
Trainings

Preparatory Steps

⚠ WARNING ⚠

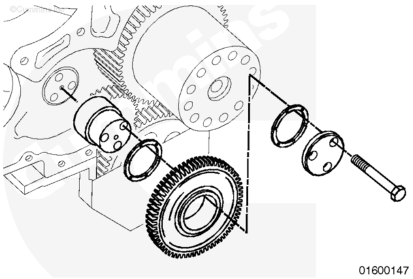
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html)

Remove

The bolt-in idler shafts have a flange that requires the shaft, idler gear, and thrust washer to be removed as an assembly.



Mark the position of the idler shaft before removing it.

Remove the three capscrews, retainer, and thrust washer from the idler gear shaft.

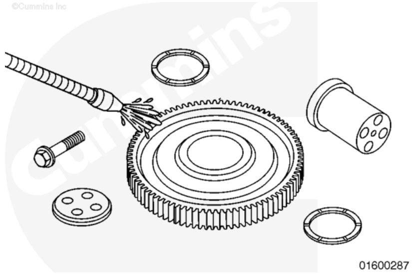
Remove the idler gear, thrust washer, and shaft as an assembly from the engine.

Remove the idler gear and thrust washer from the shaft.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



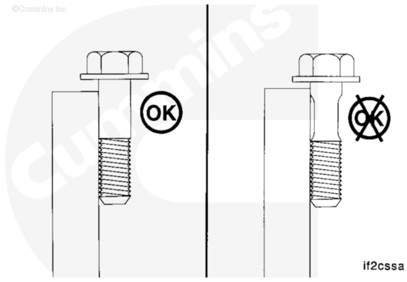
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the parts with solvent and dry with compressed air.

Use a straightedge to inspect the retaining capscrews for necking.

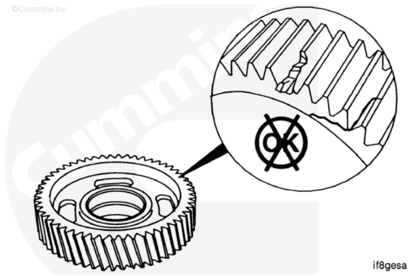
If the retaining capscrews are necked, they **must** be replaced.



Inspect the idler gear for chipped, broken, or cracked teeth.

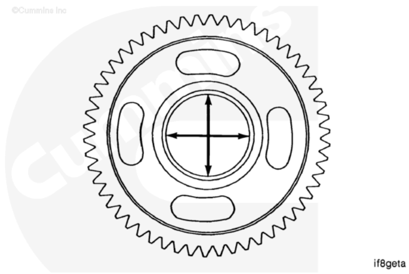
Inspect the bushing for damage.

If the idler gear bushing or teeth are damaged, the idler gear **must** be replaced.



Measure the inside diameter of the idler gear bushing.

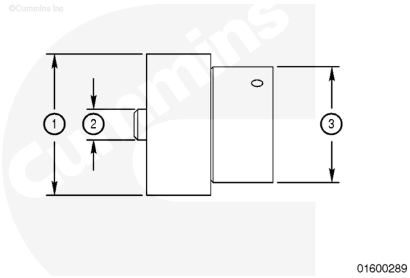
Water Pump Drive Idler Gear Inside Diameter		
mm		in
73.035	MIN	2.8753
73.065	MAX	2.8765



The bushing in the idler gear is precision bored after installation. Replace the bushing and gear as an assembly if the idler gear bushing is **not** within specification.

If the idler gear is a compound gear, it **must** be replaced as an assembly.

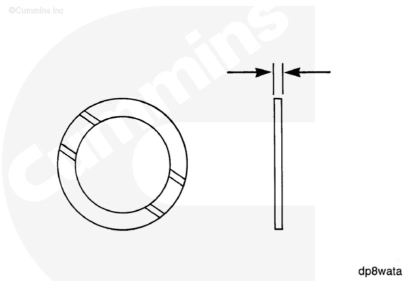
Measure the outside diameter of the idler gear shaft.



Water Pump Drive Idler Gear Shaft Outside Diameter			
	mm		in
(1)	88.75	MIN	3.494
	89.05	MAX	3.506
(2)	21.975	MIN	0.8651
	22.000	MAX	0.8661
(3)	72.92	MIN	2.871
	72.98	MAX	2.873

If the idler gear shaft is **not** within specifications, it **must** be replaced.

Check the grooved side of the thrust bearing for damage.



If the grooved side of the thrust bearing is damaged, it **must** be replaced.

Measure the thrust bearing thickness.

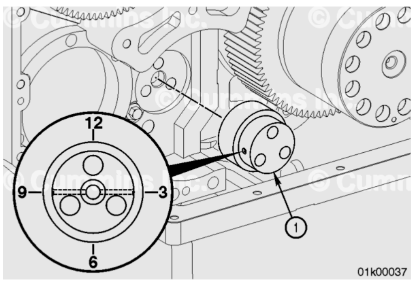
Water Pump Idler Gear Thrust Bearing Thickness		
mm		in
2.337	MIN	0.092
2.387	MAX	0.094

If the thrust bearing is **not** within specification, it **must** be replaced.

Install

⚠ CAUTION ⚠

All the holes on the outside diameter of the shaft must be in the correct position or failure from lack of lubrication will result.



For assembly purposes, the recommendation is to have the water pump idler gear part number facing away from the cylinder block.

The water pump idler shaft (1) **must** be installed so the holes on the outside diameter of the shaft are located at the 3 o'clock and 9 o'clock positions.

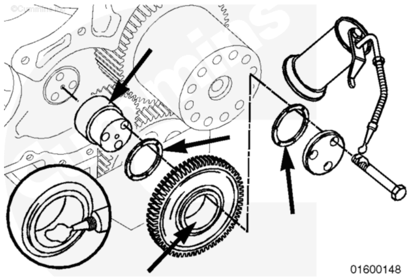
The grooves in the thrust washer must be turned toward the gear.

Use Lubriplate™ 105, or equivalent. Lubricate the gear bushing, idler shaft, and thrust washer.

Rotate the idler shaft as necessary to align the mounting capscrew holes.

Lubricate the three capscrews with clean engine oil.

Assemble the parts as shown.


01600148

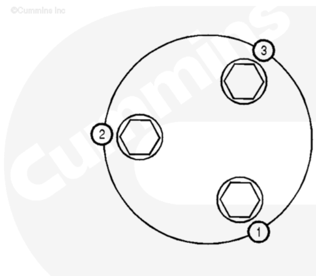
Use the three capscrews to pull the shaft into the bore.

Tighten the capscrews in the sequence shown in the illustration.

Tighten all three capscrews evenly to each torque value step.

Torque Value:

1. 80 n•m [59 ft-lb]
2. 165 n•m [122 ft-lb]
3. 280 n•m [207 ft-lb]



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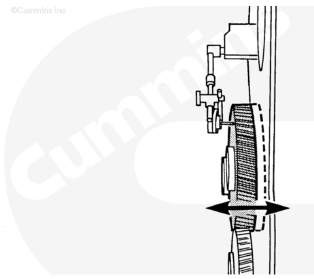
Use a dial indicator to measure the idler gear end clearance.

Water Pump Idler Gear - End Clearance		
mm		in
0.15	MIN	0.006
0.33	MAX	0.013

If the clearance is **not** within specifications, check for foreign material between the parts, or check for proper location of the thrust washer. Oversize thrust washers are available.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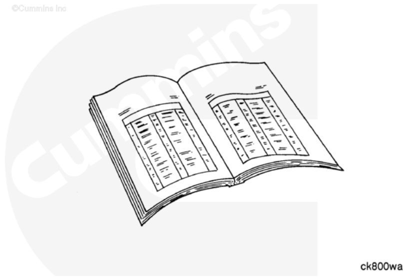
Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html)
- Operate the engine and check for leaks.

Last Modified: 30-Aug-2023